

Prompt engineering:

Prompt Engineering is the discipline of designing, structuring, and optimizing inputs to guide Large Language Models (LLMs) such as OpenAI GPT to generate accurate, context-aware, and controlled outputs.

Core Idea:

LLMs are probabilistic systems:

- They don't "know" facts
- They predict next tokens based on probability

Prompt = Control mechanism over probability distribution

LLM Sampling Parameters (Hallucination Control Layer)

These parameters control how the model selects the next word.

Temperature:

Controls randomness in token selection.

Probability distribution is **flattened or sharpened**.

- Low temperature → sharp distribution → predictable output
- High temperature → flat distribution → diverse output

Real-world Example:

- SQL generation → Temperature = 0
- Story writing → Temperature = 0.8

Top-K Sampling:

Model selects from top K most probable tokens.

- Top-K = 10 → Only 10 highest probability words considered

Advantage:

- Prevents rare/unlikely words
- Reduces hallucination

Top-P (Nucleus Sampling):

Select tokens whose cumulative probability = P

Top-P = 0.9 → choose smallest set of tokens adding to 90% probability

Prompting Techniques:

Zero-Shot Prompting

No examples given

Behavior:

Model relies on pre-trained knowledge

Example:

Explain ETL pipeline in Azure.

Pros:

- Fast
- Simple

Cons:

- May hallucinate

- Less structured

One-Shot Prompting

Provide one example

Purpose:

Guide output format

3.3 Few-Shot Prompting

Provide multiple examples

Use Case:

- Classification
- Data transformation
- Structured output

Why it works:

LLM learns pattern from examples

System, Role, and Context Prompting

System Prompt

Defines global behavior and rules

Example:

You are a senior data engineer. Answer with structured explanations.

Controls:

- Tone
- Output format
- Depth

Role Prompting

Assign a persona

Examples:

- “Act as a Data Engineer”
- “Act as an Auto Insurance Analyst”

Why important:

LLM adapts:

- Vocabulary
- Explanation style
- Domain knowledge

Contextual Prompting

Provide background information

Example:

We are using Azure Data Factory and Databricks...

Explain optimization techniques.

Impact:

- Reduces ambiguity
- Improves relevance

Advanced Prompting Techniques

Step-Back Prompting

Concept:

Ask model to first identify key principles

Flow:

1. Extract important factors
2. Then solve

Benefit:

- Reduces hallucination
- Improves reasoning

Chain of Thought (CoT)

Force step-by-step reasoning

Example:

Explain step by step how fraud detection works.

Why it works:

- Breaks complex problem
- Improves logical accuracy

Tree of Thoughts (ToT)

Explore multiple reasoning paths

Process:

1. Generate multiple solutions
2. Evaluate each
3. Select best

Used in:

- Planning systems
- Optimization problems

ReAct (Reason + Act)

Model interacts with tools

Loop:

1. Reason
2. Act (API call / tool)
3. Observe
4. Repeat

Used in:

- AI agents
- LangChain pipelines

Code Prompting

Prompting for code generation/debugging

Best Practices:

- Specify language

- Provide schema
- Define constraints

Guardrails

Guardrails are constraints that control model behavior.

Types of Guardrails

1. Input Guardrails

- Block harmful prompts

2. Output Guardrails

- Validate response

3. Policy Guardrails

- Business rules

4. Tool Guardrails

- External validation

Guardrail Prompting

Example:

Only answer using provided dataset.

If unsure, say "I don't know".

Impact:

- Reduces hallucination
- Improves trust

9. Grounding

Model answers only using trusted/internal data

Why needed?

LLMs:

- Can hallucinate
- Use outdated info

Grounding solves:

- Accuracy issues
- Trust issues

Implementation

Grounding is done using:

- RAG (Retrieval-Augmented Generation)
- Vector databases (FAISS, Pinecone, Chroma)

Real Example

Auto Insurance:

- Claims DB
- Policy DB
- Telematics

LLM answers using:

Only these sources

10. Hallucination Control Strategies

Best Practices

- Use low temperature
- Add context
- Use grounding (RAG)
- Apply guardrails
- Use step-by-step prompts